

2024.11.7 Syswonder 组会报告

Petalinux build and boot, LoongArch hvior virtio debug

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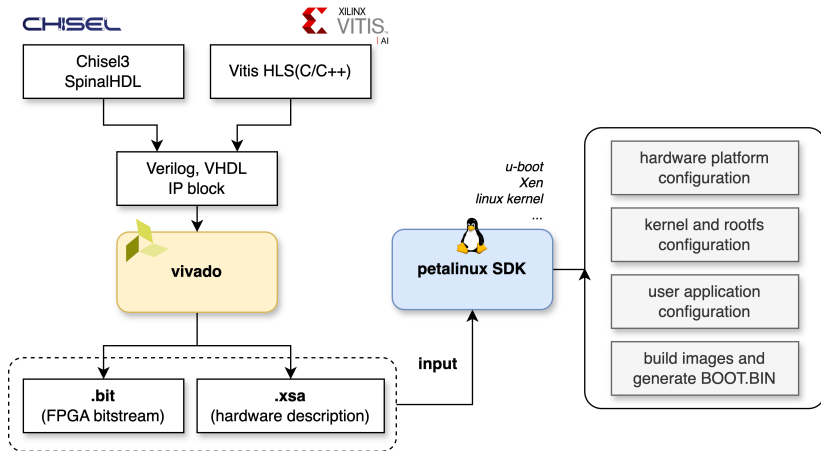
- ① Petalinux Image Creation and Boot Process
- ② LoongArch Part

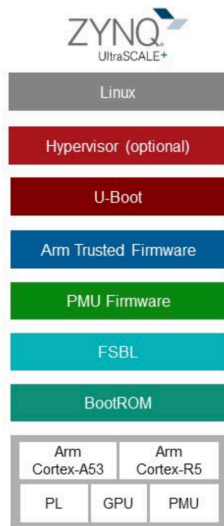
① Petalinux Image Creation and Boot Process

What about hvisor?

② LoongArch Part

Petalinux Workflow





- **BootROM:** used by CSU^a to load FSBL.
- **First Stage Boot Loader(FSBL):** configure the FPGA with bitstream(optional), load OS image, load 2nd-stage BL from non-volatile memory to memory. Finally, it resets ZCU102's A53 and R5 cores. FSBL is loaded on OCM.
- **PMU Firmware:** Xilinx's MicroBlaze processor, used for power and error management, test library(optional).
- **ARM Trusted Firmware(ATF):** a reference implementation of secure world software for ARMv8-A from ARM. Support PSCI, SMC^b, etc.

^aConfiguration Security Unit, used for secure boot, tamper monitoring, crypto, etc.

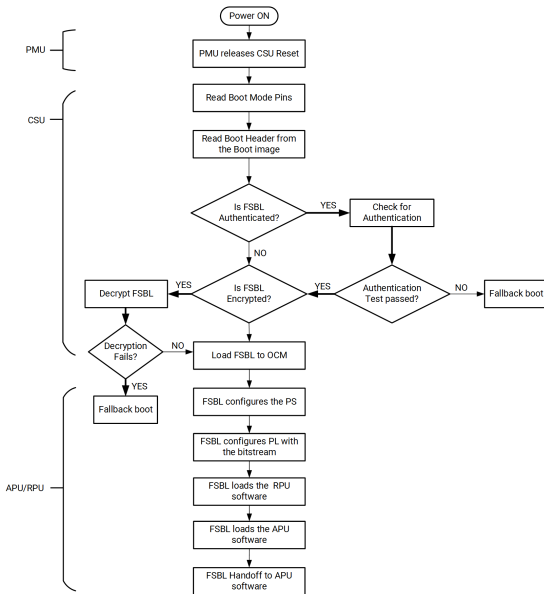
^bPower State Coordination Interface, Secure Monitor Call

Petalinux Image Files

- **image.ub**: Image(kernel image), system.dtb(device tree), and rootfs.cpio.gz.u-boot.
- **pmufw.elf**: PMU Firmware
- **zynqmp_fsbl.elf**: First Stage Boot Loader
- **bl31.elf**: ARM Trusted Firmware(Bootloader Stage 3-1¹)
- **u-boot.elf**: U-Boot program
- **boot.scr**: U-Boot script
- **BOOT.BIN**: a package of FSBL, PMUFW, ATF, U-Boot, etc. used for booting.
- BootROM loads FSBL, FSBL loads PMU Firmware, ATF, U-Boot, etc.

¹<https://chromium.googlesource.com/external/github.com/ARM-software/arm-trusted-firmware/+/-/v0.4-rc1/docs/firmware-design.md>

Boot Flow²



X14935-070717

²<https://docs.amd.com/r/en-US/ug1137-zynq-ultrascale-mpsoc-swdev/Pre-Boot-Sequence>

Xilinx XSDB and XSCT

- **XSDB**: Xilinx System Debugger. **XSCT**: Xilinx Software Command-Line Tool.
- XSCT³ is a wrapper for XSDB but with more features.
- can create platform, system, application projects, build BSPs, download apps to hardware, read hardware registers, add breakpoints, JTAG, etc.
- XSCT is deprecated and a new Xilinx Python CLI tool will replace it.

³<https://docs.amd.com/r/en-US/ug1400-vitis-embedded/Software-Command-Line-Tool>

Boot from Different Places

- **SD Card**: create a BOOT.bin and place it in SD card.
- **QSPI Flash**: create a qspi_BOOT.bin and flash it into QSPI Flash using XSCT debugger.
- **JTAG**: create a BOOT.bin and boot it using XSCT debugger.

Boot from JTAG using Petalinux

- <https://docs.amd.com/r/en-US/ug1137-zynq-ultrascale-mpsoc-swdev/USB-Boot-Mode>
- **Step 1:** Connect JTAG USB cables from Xilinx ZCU102 board to PC. Set board to JTAG boot mode.
- **Step 2:** Power on the board and wait for self-test to complete.
- **Step 3:** Run `petalinux-boot jtag -prebuilt 3 -hw_server-url <hostname:3121>`
- **Step 4:** Open `ttyUSB0` using `screen` or `termius` and enter linux's shell.

What about hvisor?

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What about hvisor?

A Possible Way?

- Boot Image Format(BIF) script. To generate customized BOOT.bin.
- using BOOTGEN program to generate BOOT.bin, but replace all linux's stuff to hvisor stuff.
- **approach:** using u-boot to place hvisor.bin, root linux, non-root linux, etc. into memory, and then run hvisor.bin using u-boot script.

What about hvisor?

Example BIF for linux kernel⁴

```
1 the_ROM_image:
2 { [bootloader, destination_cpu = a53-0] fsbl_a53.elf
3   [destination_cpu=pmu] pmu_fw.elf
4   [destination_cpu=a53-0, exception_level=el-3, trustzone] bl31
   .elf
5   [destination_cpu=a53-0, exception_level=el-2] u-boot.elf
6   [offset=0x1E40000, load=0X10000000, destination_cpu=a53-0]
   image.ub
7 }
```

- **TODO:** read BOOTGEN manual and figure out how to generate BOOT.bin for hvisor.

⁴Xilinx Bootgen User Guide, UG1283 (v2022.2) December 14, 2022

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hvisor book: LoongArch Quick Start
LoongArch hvisor debug

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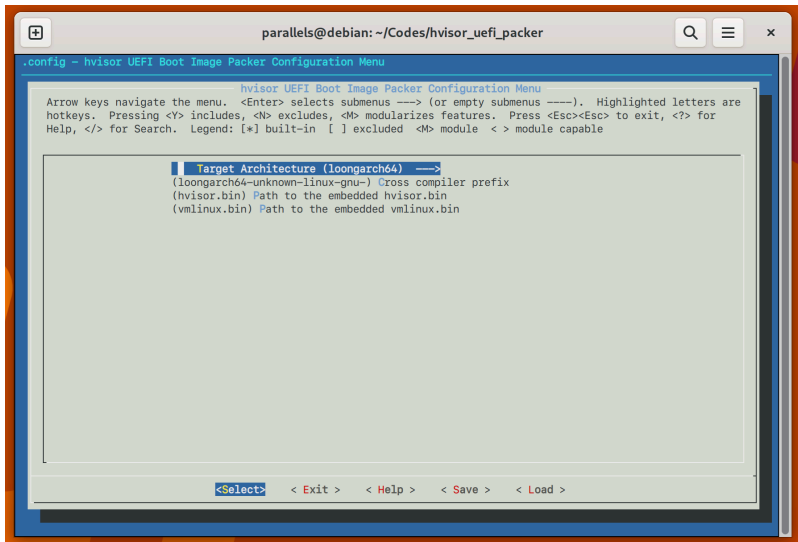
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hvisor UEFI Image Packer

- https://github.com/syswonder/hvisor_uefi_packer
- Rewrite build scripts and integrated Linux kernel's Kconfig build system for better configuration.
- Allow user to specify hvisor.bin and vmlinux.bin location using menuconfig, etc.

hvisor UEFI Image Packer Kconfig



hvisor UEFI Image Packer

- Add *Loongson 3A5000 hvisor Quick Start* section in hvisor book (<https://hvisor.syswonder.org/>).

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hvisor book: LoongArch Quick Start

LoongArch hvisor debug

Bad news

- **Problem:** new 3A5000 machine uses a new version of UEFI firmware, hvisor can boot, but root-linux failed to boot and stuck.
- the stuck appears after kernel trying to configure device tree's interrupt controllers, and then the CPU died, no output and no exception, even JTAG won't work after the stuck.
- **TODO:** try to find the cause of the stuck in the kernel boot process.

Thanks for watching!

- TODO1: read BOOTGEN manual and figure out how to generate BOOT.bin for hvisor.
- TODO2: try to find the cause of the stuck in the kernel boot process.